Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

Final Project Report

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Abstract. This report analyzes 822 monthly mean concentrations of atmospheric carbon dioxide recorded at Mauna Loa Observatory from March 1958 to August 2026. The series has a strong, accelerating trend and a stable 12-month cycle, so both a regular and a seasonal difference are required for stationarity. Four model families were compared: ARMA, ARIMA, seasonal ARIMA, and regression with autocorrelated errors. Identification from the sample ACF and PACF, information criteria and residual checks select the seasonal model ARIMA$(0,1,1) \times (0,1,1)_{12}$. Both moving-average parameters are highly significant, the model is invertible, and its residuals pass the Ljung–Box test at lags 6 through 24 and show constant variance. A mild departure from normality is traced to five isolated months and does not affect the estimates. Across 276 rolling out-of-sample forecasts it matches Holt–Winters smoothing and beats the other model families. The model forecasts that monthly CO_2 will first exceed 440 ppm in April 2030.

Keywords: seasonal ARIMA; ARMA; regression with autocorrelated errors; Holt–Winters; Mauna Loa.

1. Topic and Motivation

Carbon dioxide is the most important long-lived greenhouse gas, and the Mauna Loa record, started by Charles D. Keeling in 1958, is the longest continuous direct measurement of it (Keeling, 1960). I chose it because its trend, seasonal cycle and short-term dependence allow a full range of time series methods to be applied to data of clear scientific importance, and because it raises questions the methods can answer: how fast is CO_2 rising, is the rise accelerating, and what levels should be expected by 2030?

2. Description of the Series

The data are the monthly mean CO_2 mole fractions, in parts per million (ppm), published by the NOAA Global Monitoring Laboratory (Lan and Keeling, 2026). The series X_t contains $n = 822$ consecutive months from March 1958 to August 2026, with no missing values. Concentration rose from 315.71 ppm in March 1958 to 427.55 ppm in August 2026, an increase of 111.8 ppm (35%). The mean is 361.36 ppm (SD 33.46 ppm); the maximum, 432.34 ppm, occurred in May 2026. The average 12-month increase over the whole record is 1.68 ppm per year, but this rate has more than tripled, from 0.86 ppm/yr in the 1960s to 2.40 ppm/yr in the 2010s and 2.54 ppm/yr in the 2020s. The largest 12-month increases occurred in March 2024 (4.39 ppm) and April 2016 (4.25 ppm), both El Niño years.

A classical additive decomposition with a 2×12 moving-average filter (Brockwell and Davis, 2016) attributes 96.6% of the variance of X_t to the trend. The seasonal component peaks in May (+3.04 ppm) and reaches its minimum in October (-3.25 ppm), an amplitude of 6.29 ppm. This reflects uptake by Northern Hemisphere vegetation in the growing season; the irregular component is small (SD 0.30 ppm).

3. Stationarity of the Series

Time plot. The first step was the time plot, which shows two clear features: a persistent upward trend, and a regular 12-month seasonal cycle that repeats every year with nearly the same shape. The mean of X_t therefore depends on t, and the series is non-stationary (Brockwell and Davis, 2016).

Transformation. A log or Box–Cox transformation is warranted when variability grows with the level (Brockwell and Davis, 2016). The within-year range of the detrended series stays between 5.3 and 7.6 ppm (6.0 ppm in the 1960s, 7.0 ppm in the 2010s) while the level rose 35%, so the additive form is adequate and no transformation is used.

Sample ACF. The sample ACF confirms the non-stationarity: $\hat{\rho}(1) = 0.996$, and the ACF is still 0.865 at lag 36. After a first difference, ∇X_t, the ACF at the seasonal lags 12, 24 and 36 remains 0.925, 0.910 and 0.894. This failure to decay shows that the seasonal pattern is also non-stationary, so a seasonal difference $\nabla_{12} = 1 - B^{12}$ is required. The seasonally differenced series $\nabla_{12} X_t$ is the 12-month change in concentration. Its ACF still decays slowly (0.856, 0.792 and 0.745 at lags 1–3), and its level rises steadily over time as the growth rate accelerates, so it does not have a constant mean.

Augmented Dickey–Fuller tests. Because seasonal swings mimic mean reversion, the ADF test (Dickey and Fuller, 1979) was applied to the seasonally adjusted series. With a single mean, τ ranges from 3.57 to 4.85, far above the 5% critical value of -2.86; with a trend, τ ranges from -0.88 to -0.62, against a critical value of -3.41. Adding 13 augmentation lags to the raw series gives $\tau = -0.33$. A unit root is never rejected. For $\nabla_{12} X_t$ the test rejects (τ from -7.91 to -5.37), but an AR(1)-plus-drift model for $\nabla_{12} X_t$ estimates $\hat{\phi} = 0.9993$ (SE 0.0010), practically a unit root. For the doubly differenced series $\nabla \nabla_{12} X_t$, the test rejects decisively at every lag (τ from -37.89 to -22.05). I therefore model $W_t = (1 - B)(1 - B^{12}) X_t$, which has 809 observations, as a stationary series.

Estimation versus differencing. As an alternative to differencing, I estimated the deterministic components by regression on a quadratic trend and 11 monthly dummy variables. The fit is excellent ($R^2 = 0.9994$, residual SD 0.82 ppm), and the positive quadratic coefficient confirms that the trend is accelerating. The residuals, however, are far from white noise: their lag-1 autocorrelation is 0.915, $Q(12) = 5991$, and the Durbin–Watson statistic is $D = 0.17$, far below the value of 2 expected for independent errors. Deviations from the fixed polynomial persist for years, so differencing is the more appropriate treatment; a regression with ARMA errors is evaluated as a competing model in Section 4.

4. Model Selection and Model-Building Procedure

The model was built with the Box–Jenkins procedure (Box et al., 2015) (plot, transform, identify, estimate, diagnose, forecast), with $d = D = 1$ and $T = 12$ fixed in Section 3.

White-noise test. The Ljung–Box test (Ljung and Box, 1978) for W_t gives $Q(6) = 69.99$, $Q(12) = 283.32$, $Q(18) = 295.43$ and $Q(24) = 301.19$ (all $p < 0.001$), so W_t is not white noise and must be modeled.

Identification. The sample ACF of W_t has a significant spike at lag 1 (-0.280) and a large spike at lag 12 (-0.471), with smaller satellite values at lags 11 (0.185) and 13 (0.042). Almost all other

autocorrelations lie within the $\pm 1.96/\sqrt{809} = \pm 0.069$ bounds. The sample PACF, by contrast, tails off: at lags 1–3 it is -0.280, -0.146 and -0.135, and at the seasonal lags 12, 24 and 36 it is -0.378, -0.260 and -0.163. An ACF that cuts off together with a PACF that tails off is the signature of moving-average behavior (Brockwell and Davis, 2016). Here that behavior appears at both the non-seasonal lag 1 and the seasonal lag 12. This points to the multiplicative seasonal ARIMA model (Box et al., 2015):

$$(1-B)(1-B^{12})X_t = (1-\theta_1 B)(1-\Theta_1 B^{12})Z_t, \qquad Z_t \sim \mathrm{WN}(0,\sigma_Z^2), \tag{1}$$

that is, ARIMA$(0,1,1)\times(0,1,1)_{12}$. Its right-hand side is an MA(13) with only two free parameters, which explains the small autocorrelations at lags 11 and 13.

Estimation and selection. Nine seasonal ARIMA candidates were estimated by maximum likelihood (ML) and compared with AIC and SBC (Schwarz, 1978). ARIMA$(0,1,1)\times(0,1,1)_{12}$ attains the lowest SBC (465.39); ARIMA$(1,1,1)\times(0,1,1)_{12}$ has a lower AIC (447.58 versus 451.30) but a higher SBC (466.37). Pure AR alternatives fit worse, as expected when the PACF tails off rather than cutting off: ARIMA$(1,1,0)\times(0,1,1)_{12}$ and ARIMA$(2,1,0)\times(0,1,1)_{12}$ have SBC 486.92 and 479.42. Adding seasonal AR terms raises the SBC by at least 6.7, and ARIMA$(0,1,1)\times(1,1,0)_{12}$ is clearly inadequate (SBC 683.43; residuals fail the Ljung–Box test). Because SBC penalizes extra parameters more heavily in large samples, the two-parameter model was selected. Its implied autocorrelations, $\rho(1) = -0.338$, $\rho(12) = -0.495$ and $\rho(11) = \rho(13) = 0.167$, closely match the sample values above.

Comparison of the four model families. Four model families were fitted, each with its order chosen by SBC. (a) *ARMA*: the best non-seasonal model for W_t, ARMA(3,2), leaves a residual autocorrelation of -0.409 at lag 12 (Ljung–Box $p < 0.001$), because a non-seasonal ARMA cannot represent the seasonal MA structure. (b) *ARIMA*: the best ARIMA$(p,1,q)$ with drift, ARIMA(4,1,4), has four AR roots of modulus 1.000–1.001 with periods of 6 and 12 months. It imitates a seasonal difference with unit roots, and its residuals remain correlated at lag 12 (0.180, $p < 0.001$). (c) *Regression with autocorrelated errors*: the quadratic trend and monthly dummies with ARMA(2,1) errors give an AR root of 1.02, almost a unit root, and residual correlation of 0.138 at lag 12 ($Q(12)$ $p < 0.001$). The standard remedy is then to difference both sides at lags 1 and 12, which removes the regressors and leads back to a seasonal ARIMA model. (d) *Seasonal ARIMA*: only ARIMA$(0,1,1)\times(0,1,1)_{12}$ leaves white-noise residuals (lag-12 autocorrelation -0.005, $Q(12)$ $p = 0.464$). The seasonal ARIMA family is therefore required, and the selected model is itself an ARMA model for W_t, namely an MA(13) restricted to two parameters.

5. Hypothesis Tests and Confidence Intervals

The ML estimates are $\hat\theta_1 = 0.3885$ (SE 0.0367) and $\hat\Theta_1 = 0.8700$ (SE 0.0170), with $\hat\sigma_Z^2 = 0.1012$, so $\hat\sigma_Z = 0.318$ ppm. The log-likelihood is -222.65. For $H_0 : \theta_1 = 0$, $z = 10.59$ ($p < 0.001$), with 95% confidence interval $[0.317, 0.460]$. For $H_0 : \Theta_1 = 0$, $z = 51.28$ ($p < 0.001$), with 95% interval $[0.837, 0.903]$. Both terms are therefore essential. Conditional least squares gives similar values, $\hat\theta_1 = 0.4172$ (SE 0.0360) and $\hat\Theta_1 = 0.8116$ (SE 0.0194). This shows that the conclusions do not depend on the estimation method.

The root of $1-\hat\theta_1 B$ is $1/\hat\theta_1 = 2.57$, and the roots of $1-\hat\Theta_1 B^{12}$ have modulus $\hat\Theta_1^{-1/12} = 1.012$. Both lie outside the unit circle, so W_t is *invertible*, and with no AR part it is also causal; the model can therefore be used for forecasting. The interval for Θ_1 lies entirely below 1; $\Theta_1 = 1$ would

indicate a deterministic seasonal pattern, so seasonal differencing is justified. The large $\hat{\Theta}_1$ means the seasonal pattern changes only slowly from year to year, and $\hat{\theta}_1$ means about 39% of each month's shock is reversed the following month.

6. Model Adequacy Checking and Remedial Methods

Independence. The residuals pass the Ljung–Box test at all four lags examined: $Q(6) = 9.05$ ($p = 0.060$), $Q(12) = 9.74$ ($p = 0.464$), $Q(18) = 14.43$ ($p = 0.567$) and $Q(24) = 20.83$ ($p = 0.532$), with degrees of freedom reduced by the two estimated parameters. Only one of the first 36 residual autocorrelations lies outside ± 0.069: lag 3, at -0.081. About 1.8 such exceedances are expected by chance among 36 lags of white noise.

Constant variance. The Ljung–Box test on the squared residuals gives $Q(12) = 12.40$ ($p = 0.414$) and $Q(24) = 35.36$ ($p = 0.063$). The residual variance is similar in the first and second halves of the sample (0.091 and 0.107). There is no evidence that the variance changes over time.

Normality. The residuals are nearly symmetric (skewness 0.11, excess kurtosis 0.35), but three tests reject normality at 5%: Shapiro–Wilk $W = 0.9963$ ($p = 0.048$), Anderson–Darling ($p = 0.038$) and Jarque–Bera ($p = 0.006$). Five months have standardized residuals beyond ± 3: March 1959 (-3.46), March 1977 (3.31), September 1986 (3.11), April 2016 (3.97) and March 2022 (-3.53). The April 2016 value coincides with the unusually rapid CO_2 growth during the 2015–16 El Niño.

Remedial methods.

(i) *Lag-3 autocorrelation.* To remove this correlation, I added an AR(1) term. ARIMA$(1,1,1) \times (0,1,1)_{12}$ has a significant $\hat{\phi}_1 = 0.215$ (SE 0.084, $p = 0.010$) and reduces the lag-3 residual autocorrelation to -0.042. Its AR and MA roots (4.65 and 1.74) differ, so there is no parameter redundancy, but it raises the SBC and, in the hold-out of Section 7, forecast less accurately (RMSE 1.168 versus 1.110) with only 83% interval coverage against 97%. The parsimonious model was retained.

(ii) *Non-normality.* The five outlying months were treated as missing and the model was re-estimated by exact likelihood (Kalman filter). The estimates were almost unchanged ($\hat{\theta}_1 = 0.3838$, $\hat{\Theta}_1 = 0.8689$), and normality was no longer rejected (Shapiro–Wilk $p = 0.211$, Anderson–Darling $p = 0.101$, Jarque–Bera $p = 0.094$). The residuals still passed the Ljung–Box tests ($p \geq 0.092$). The non-normality is thus caused by a handful of genuine but unusual months. With $n = 822$ the ML estimates are approximately normal by large-sample theory, so the tests and intervals above remain valid, and the full-data model is kept because there is no reason to discard real observations.

Because the residual variance is stable, no variance-stabilizing transformation is needed, and the fitted ARIMA$(0,1,1) \times (0,1,1)_{12}$ model is adequate.

7. Forecasting

Rolling-origin comparison. Each method was re-estimated at 23 forecast origins (every August from 2000 to 2022) and forecast the next 12 months, giving 276 out-of-sample forecasts per method. The seasonal ARIMA model achieved RMSE 0.552 ppm, with 95% interval coverage of 96.0%. It was essentially tied with Holt–Winters exponential smoothing (Winters, 1960) (0.543 ppm, 96.0%), and more accurate than regression with ARMA errors (0.580 ppm, 93.8%), the ARMA model for W_t (0.677 ppm) and non-seasonal ARIMA (0.697 ppm). On the full data, Holt–Winters estimates

$\hat{\alpha} = 0.534$, $\hat{\beta} = 0.014$ and $\hat{\gamma} = 0.299$, with a current slope of 0.203 ppm per month.

Recent hold-out. Fitted to data through August 2023 and used to forecast September 2023 to August 2026, the seasonal ARIMA model had RMSE 1.110 ppm, MAPE 0.24% and 97.2% interval coverage. Holt–Winters had RMSE 0.981 ppm. The trend-and-dummies regression had RMSE 1.491 ppm with only 52.8% coverage, as its intervals ignore the residual autocorrelation, and the seasonal naive method 6.448 ppm. All methods under-predicted 2024, when the annual mean rose by a record 3.52 ppm during the 2023–24 El Niño. The ARMA model without a seasonal MA term had the lowest error in this single window (0.655 ppm) only because it carries forward the most recent annual increase, which happened to anticipate the 2024 surge. Over the 23 rolling origins it was clearly worse.

Forecasts to 2030. The model estimated on all 822 months gives a forecast for September 2026 of 426.16 ppm, with 95% prediction interval [425.53, 426.78]. For the May 2027 seasonal peak it gives 434.36 ppm [433.11, 435.60]. The forecast first exceeds 440 ppm in April 2030 (440.84 ppm [437.78, 443.89]) and peaks at 441.59 ppm [438.49, 444.69] in May 2030. The forecast for December 2030 is 439.03 ppm [435.59, 442.47]. Forecast annual means rise from 431.44 ppm (2027) to 438.67 ppm (2030), about 2.4 ppm per year, and the 95% interval widens from 1.25 ppm to 6.88 ppm, reflecting the accumulated uncertainty of a doubly integrated process. Holt–Winters gives 439.16 ppm for December 2030, within 0.13 ppm of the ARIMA forecast.

8. Conclusions

Monthly CO_2 at Mauna Loa is a non-stationary series with an accelerating stochastic trend and a stable additive seasonal cycle. Regular and seasonal differencing produce a stationary series. Of the four model families compared (ARMA, ARIMA, seasonal ARIMA, and regression with autocorrelated errors), only the seasonal ARIMA leaves white-noise residuals and gives the most accurate rolling forecasts, and the ACF/PACF and information criteria point to the parsimonious ARIMA$(0,1,1) \times (0,1,1)_{12}$. The fitted model is invertible, its parameters are precise, its residuals are uncorrelated with constant variance, and its inferences are robust to the few outlying months. CO_2 is not only increasing but increasing faster, from under 1 ppm per year in the 1960s to about 2.5 ppm per year today. The model projects that concentrations will cross 440 ppm in spring 2030. Two limitations remain. The model is univariate, so it cannot anticipate El Niño-driven accelerations such as those of 2016 and 2024, and it assumes that the current emissions-driven growth continues. Adding climate indices or emissions data as explanatory variables is a natural extension.

Supplementary material. Figures, tables and R 4.3.3 code reproducing every number are provided separately.

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